Processing Sequential Files

Run of SeqWrite

```
Enter student details using template below.
NNNNNNNSSSSSSSSIIYYMMDDCCCCGGGGS
9456789COUGHLANMS580812LM510598M
NNNNNNNSSSSSSSSIIYYMMDDCCCCGGGGS
9367892RYAN    TG521210LM601222F
NNNNNNNSSSSSSSSIIYYMMDDCCCCGGGGS
9368934WILSON  HR520323LM610786M
NNNNNNNSSSSSSSSIIYYMMDDCCCCGGGGS
CarriageReturn
```

```
     $ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.  SeqWrite.
AUTHOR.  Michael Coughlan.

ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
   SELECT StudentFile ASSIGN TO "STUDENTS.DAT"
            ORGANIZATION IS LINE SEQUENTIAL.

DATA DIVISION.
FILE SECTION.
FD StudentFile.
01 StudentDetails.
  02  StudentId       PIC 9(7).
  02  StudentName.
      03 Surname      PIC X(8).
      03 Initials     PIC XX.
  02  DateOfBirth.
      03 YOBirth      PIC 9(2).
      03 MOBirth      PIC 9(2).
      03 DOBirth      PIC 9(2).
  02  CourseCode      PIC X(4).
  02  Grant           PIC 9(4).
  02  Gender          PIC X.
```

```
PROCEDURE DIVISION.
Begin.
        OPEN OUTPUT StudentFile
        DISPLAY "Enter student details using template below.  Press CR to
    end.". PERFORM GetStudentDetails
        PERFORM UNTIL StudentDetails = SPACES
            WRITE StudentDetails
            PERFORM GetStudentDetails
        END-PERFORM
        CLOSE StudentFile
        STOP RUN.

GetStudentDetails.
        DISPLAY "NNNNNNNSSSSSSSSIIYYMMDDCCCCGGGGS".
        ACCEPT  StudentDetails.
```

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.  SeqRead.
AUTHOR.  Michael Coughlan.

ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
   SELECT StudentFile ASSIGN TO "STUDENTS.DAT"
            ORGANIZATION IS LINE SEQUENTIAL.

DATA DIVISION.
FILE SECTION.
FD StudentFile.
01 StudentDetails.
  02  StudentId       PIC 9(7).
  02  StudentName.
      03 Surname      PIC X(8).
      03 Initials     PIC XX.
  02  DateOfBirth.
      03 YOBirth      PIC 9(2).
      03 MOBirth      PIC 9(2).
      03 DOBirth      PIC 9(2).
  02  CourseCode      PIC X(4).
  02  Grant           PIC 9(4).
  02  Gender          PIC X.
```

```
PROCEDURE DIVISION.
Begin.
  OPEN INPUT StudentFile
  READ StudentFile
     AT END MOVE HIGH-VALUES TO StudentDetails
  END-READ
  PERFORM UNTIL StudentDetails = HIGH-VALUES
     DISPLAY StudentId SPACE StudentName SPACE CourseCode
     READ StudentFile
        AT END MOVE HIGH-VALUES TO StudentDetails
     END-READ
  END-PERFORM
  CLOSE StudentFile
  STOP RUN.
```

RUN OF SeqRead

```
9456789 COUGHLANMS LM51

9367892 RYAN    TG LM60

9368934 WILSON  HR LM61
```

Organization and Access

Two important characteristics of files are

- **DATA ORGANIZATION**
- **METHOD OF ACCESS**

Data organization refers to the way the records of the file are organized on the backing storage device. COBOL recognizes three main file organizations;

- Sequential - Records org

The method of access refers to the way in which records are accessed.

Sequential Organization

- The simplest COBOL file organization is **Sequential**.
- In a Sequential file the records are arranged **serially,** one after another, like cards in a dealing shoe.
- In a Sequential file the only way to access any particular record is to;

  Start at the firs
  succeeding records until yo

- Sequential files may be

  Ordered

  Unordered

- The ordering of the records in a file has a significant impact on the way in which it is processed and the processing that can be done on it.

Ordered and Unordered Files

Ordered File

Ordered File	
RecordA	RecordM
RecordB	RecordH
RecordG	RecordB
RecordH	RecordN
RecordK	RecordA
RecordM	RecordK
RecordN	RecordG

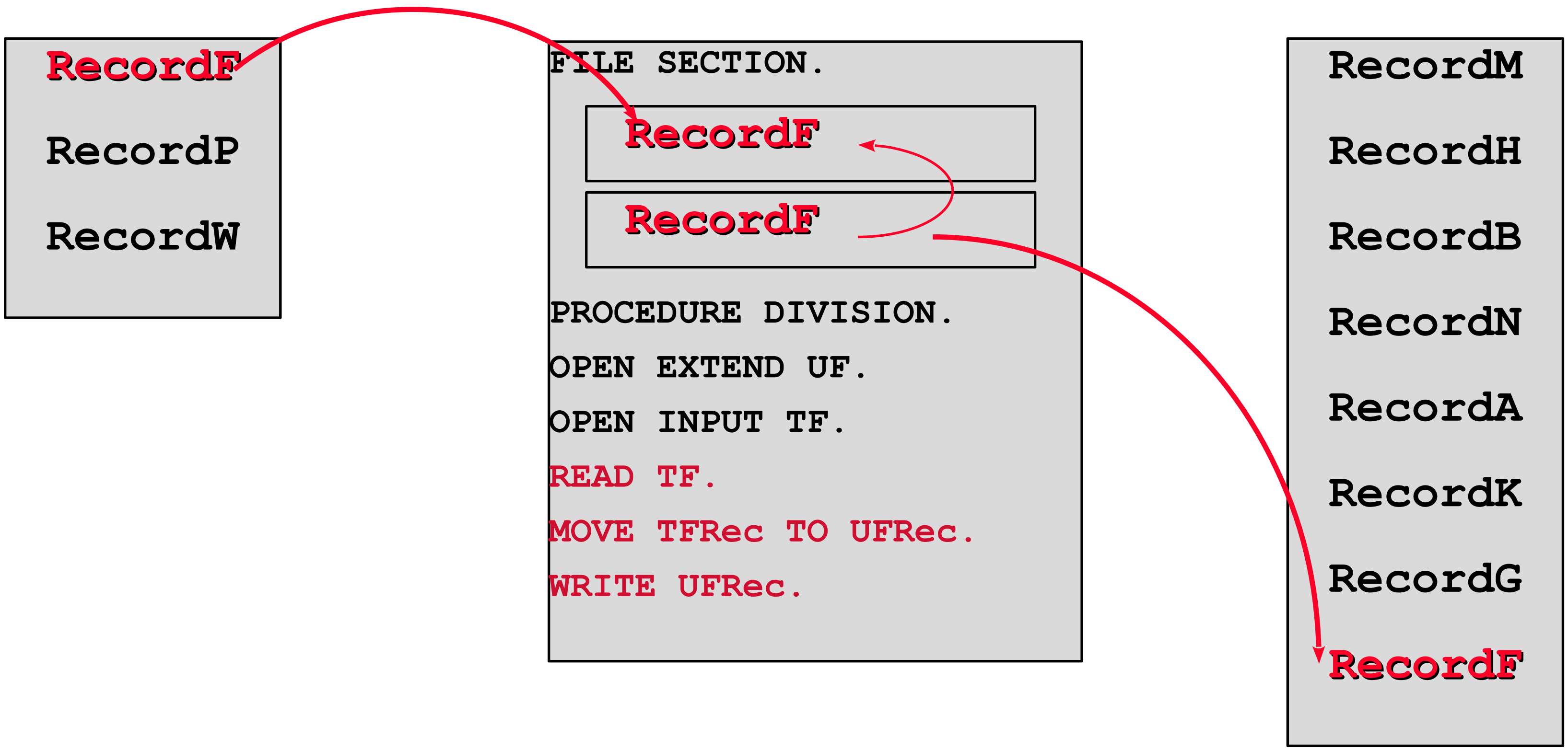
Adding records to unordered files

RecordF
RecordP
RecordW

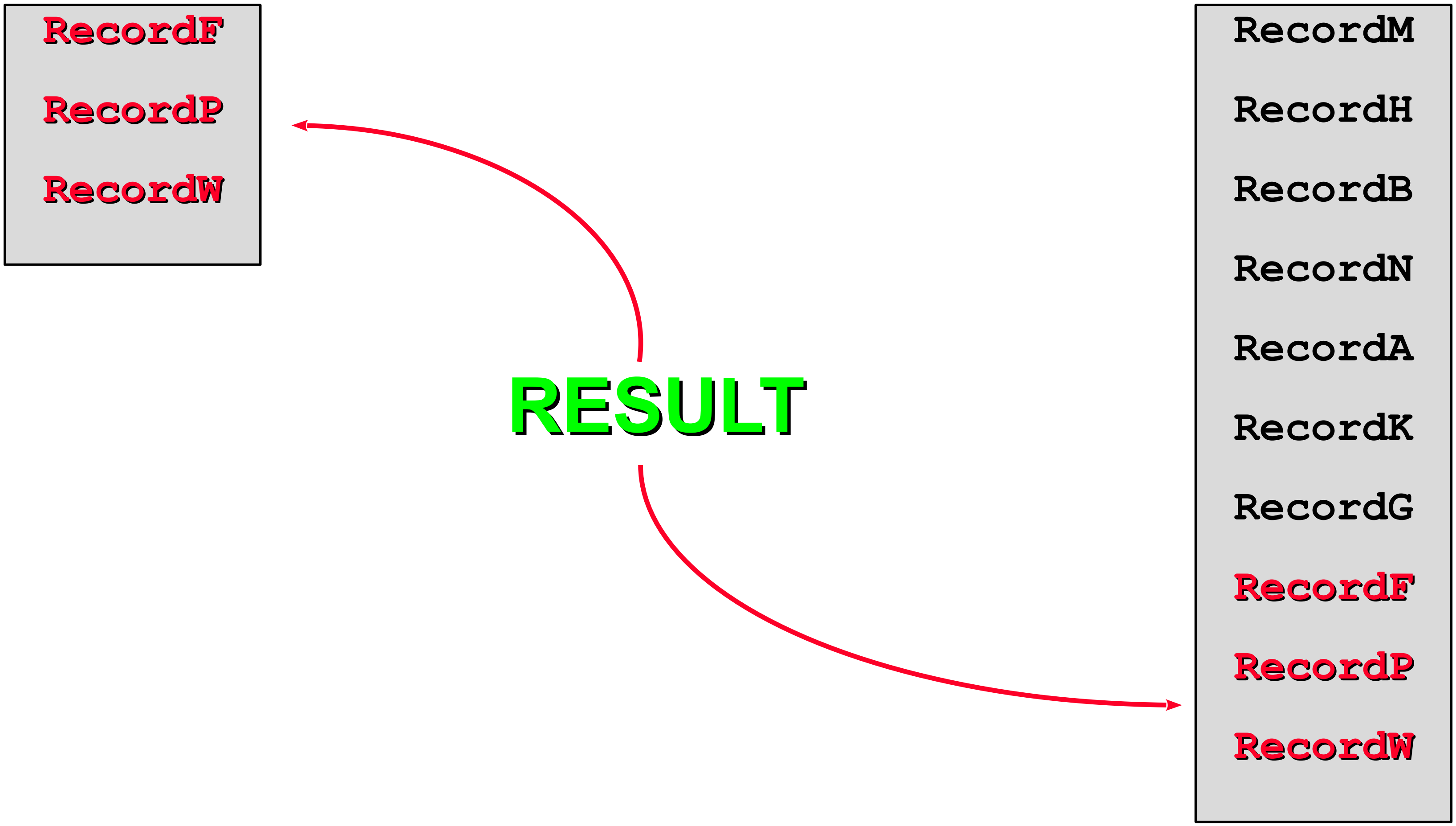
```
FILE SECTION.
    TFRec
    UFRec
PROCEDURE DIVISION.
OPEN EXTEND UF.
OPEN INPUT TF.
READ TF.
MOVE TFRec TO UFRec.
WRITE UFRec.
```

RecordM
RecordH
RecordB
RecordN
RecordA
RecordK
RecordG

Adding records to unordered files



Adding records to unordered files



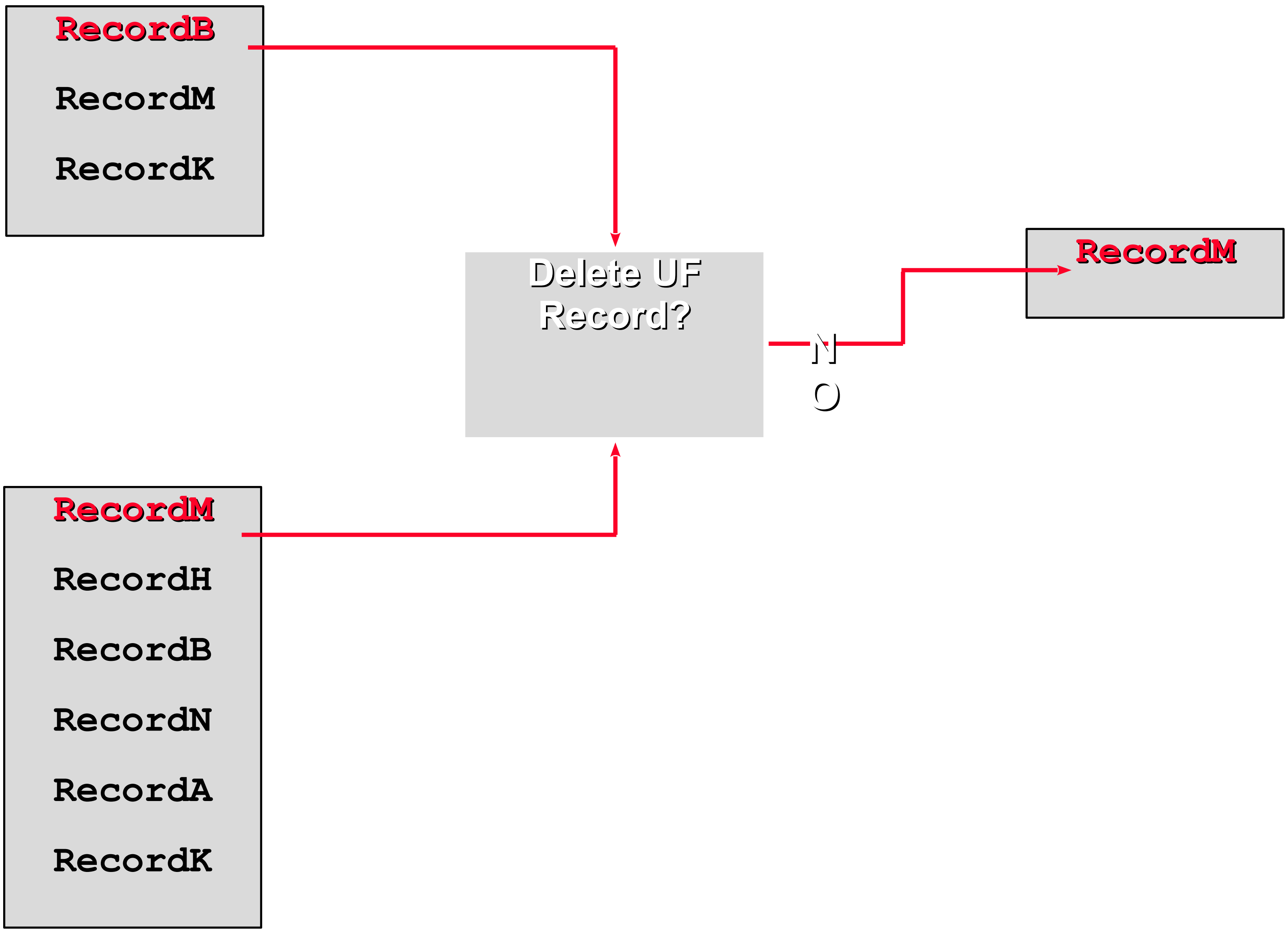
Problems with Unordered Sequential Files

- It is easy to **add** records to an unordered Sequential file.
- But it is not really possible to **delete** records from an unordered Sequential file.
- And it is not feasible to **update** records in an unordered Sequential file

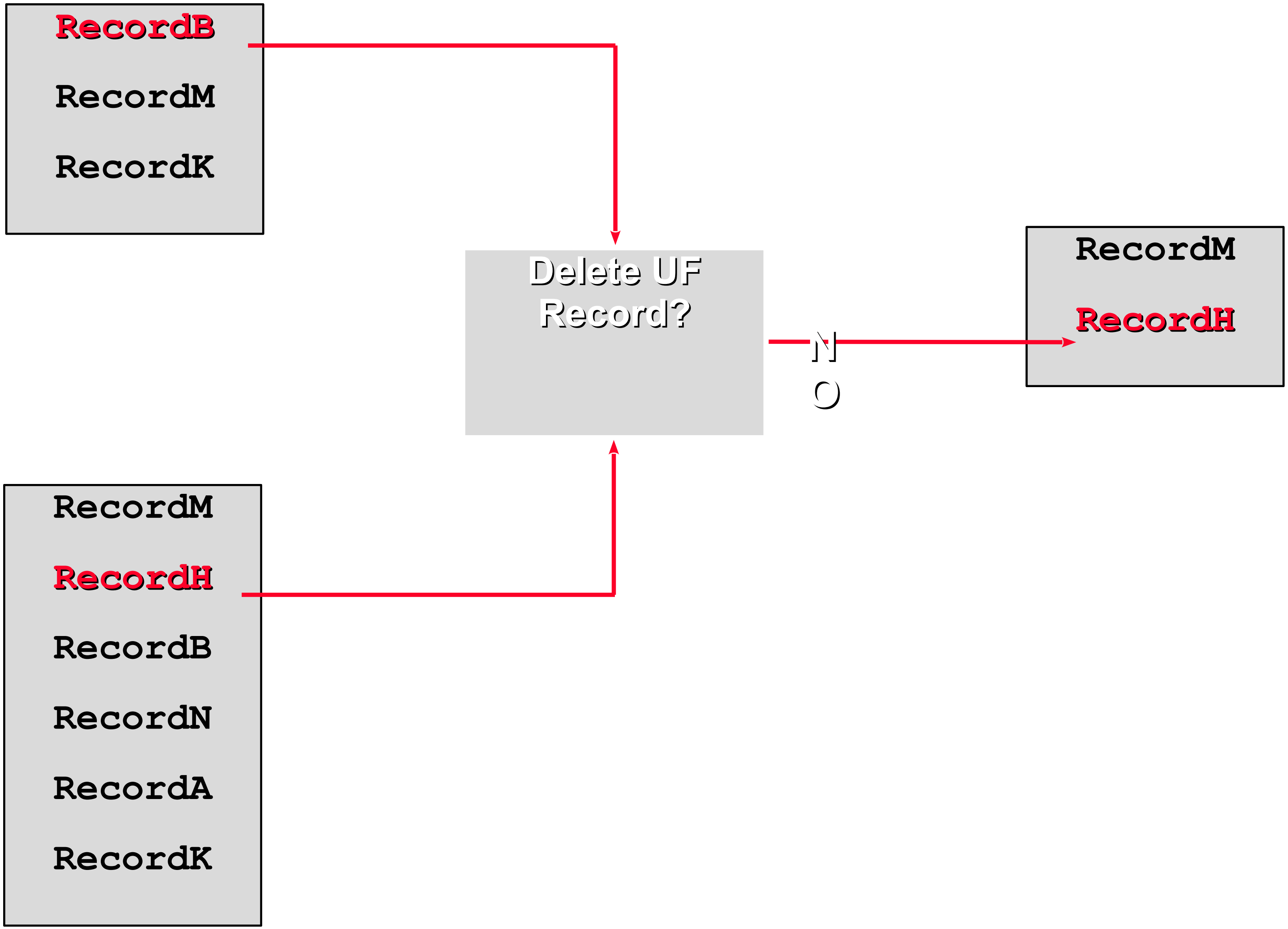
Problems with Unordered Sequential Files

- Records in a Sequential file can not be deleted or updated "in situ".
- The only way to delete Sequential file records is to create a new file which does not contain them.
- The only way to update records in a Sequential File is to create a new file which contains the updated records.
- Because both these operations rely on record matching they do not work for unordered Sequential files.
- Why?

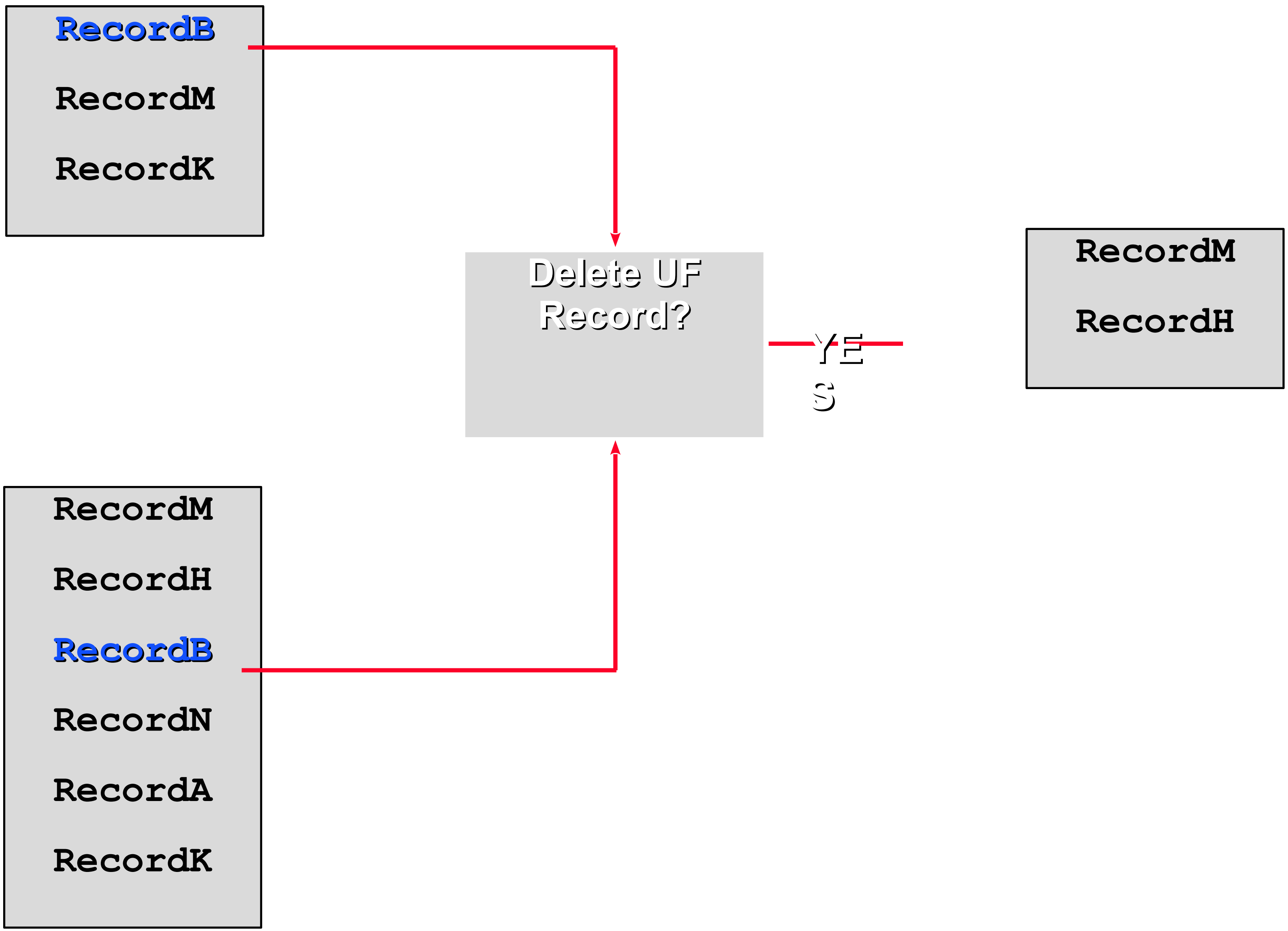
Deleting records from unordered files?



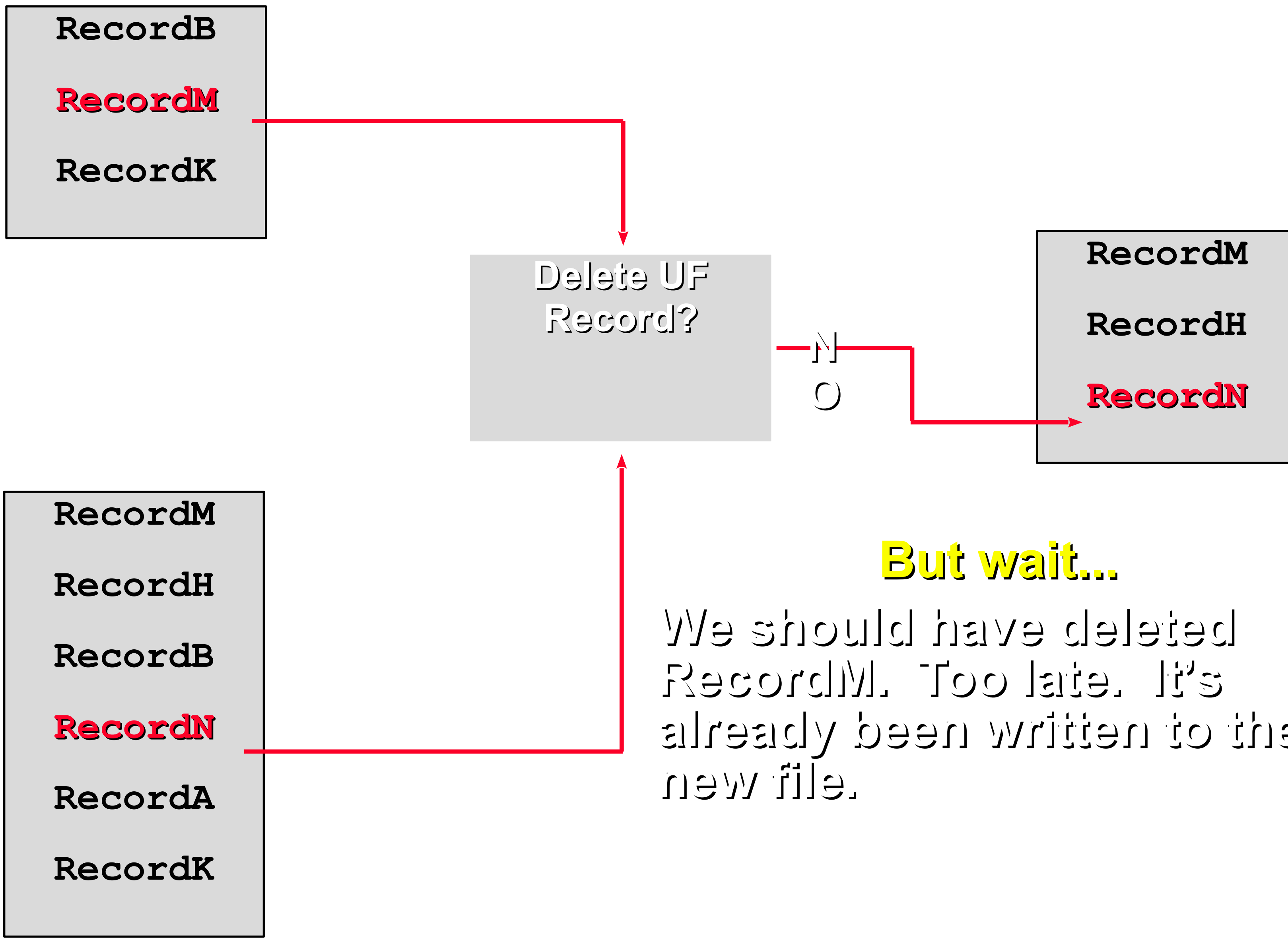
Deleting records from unordered files?



Deleting records from unordered files?



Deleting records from unordered files?



But wait...

We should have deleted RecordM. Too late. It's already been written to the new file.

Deleting records from an ordered file

RecordB
RecordK
RecordM

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

```
FILE SECTION.
    TFRec
    OFRec
    NFRec
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey NOT = OFKey
      MOVE OFRec TO NFRec
      WRITE NFRec
      READ OF
ELSE
      READ TF
      READ OF
END-IF.
```

Deleting records from an ordered file

RecordB
RecordK
RecordM

```
FILE SECTION.
    RecordB
    RecordA
    RecordA
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF.
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFRec NOT = OFRec
      MOVE OFRec TO NFRec
      WRITE NFRec
      READ OF
ELSE
      READ TF
      READ OF
END-IF.
```

RecordA

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

Problem !!

How can we recognize which record we want to delete?

By its Key Field

Deleting records from an ordered file

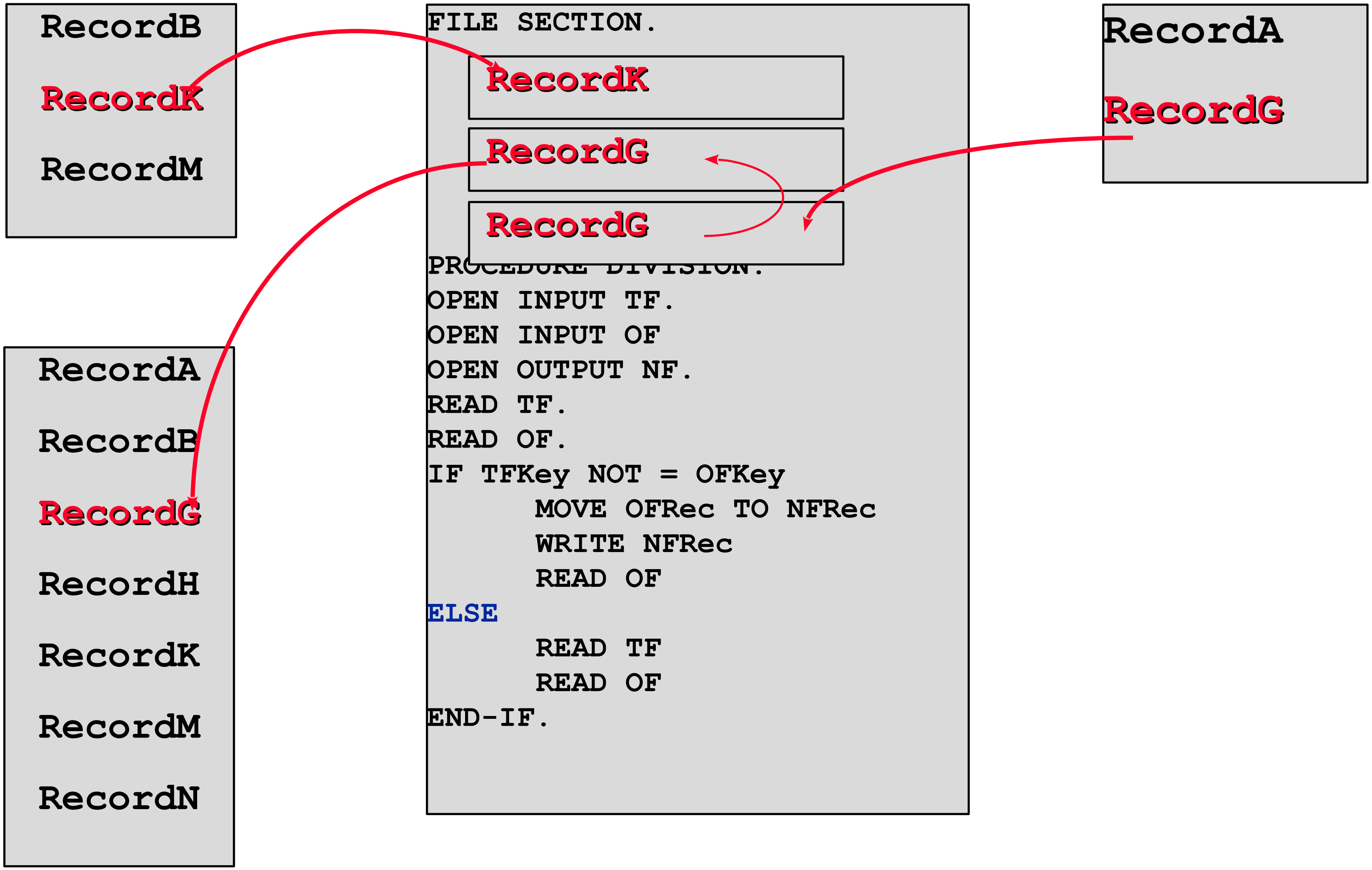
RecordB
RecordK
RecordM

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

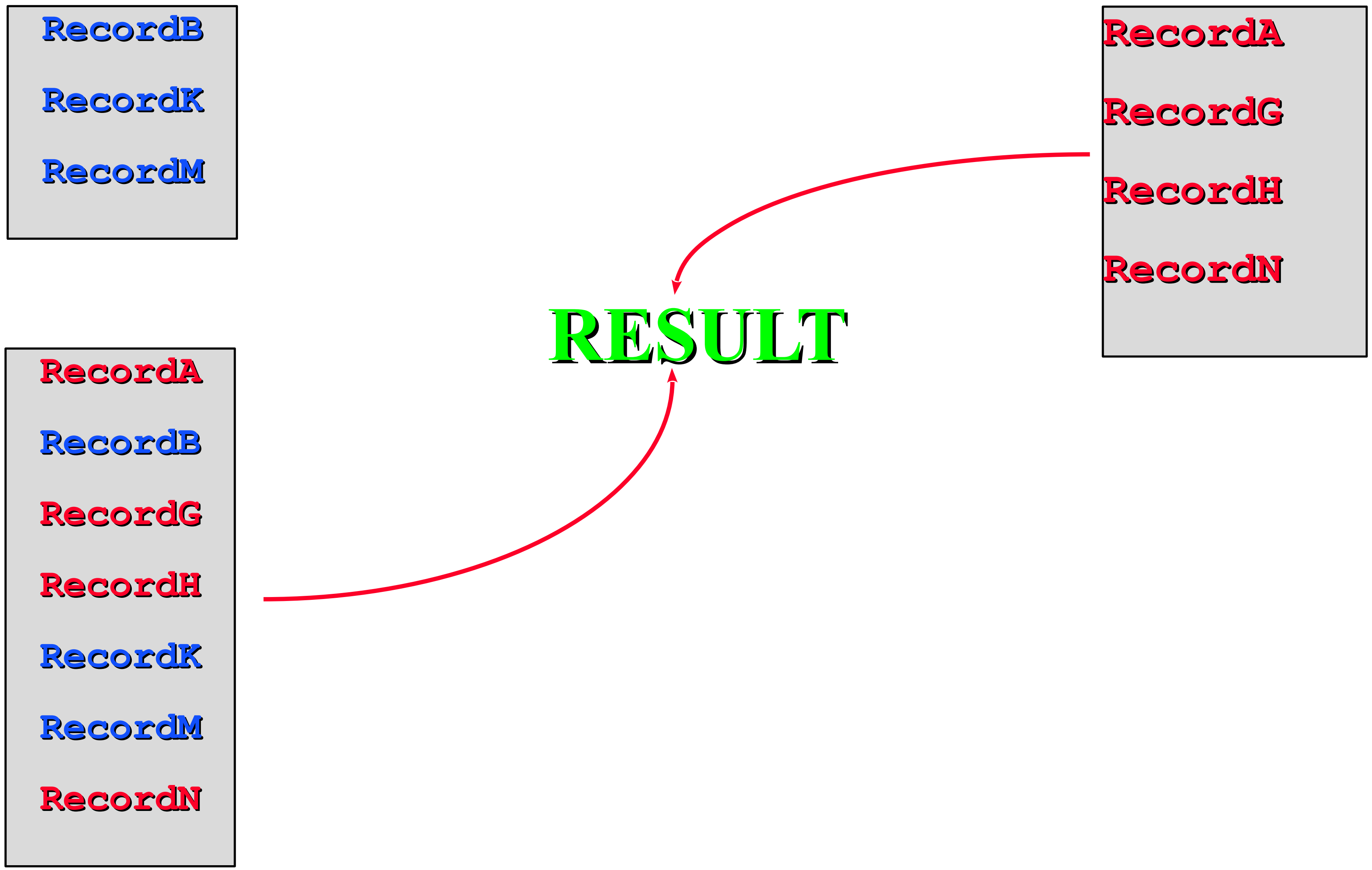
```
FILE SECTION.
    RecordB
    RecordB
    RecordA
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey NOT = OFKey
      MOVE OFRec TO NFRec
      WRITE NFRec
      READ OF
ELSE
      READ TF
      READ OF
END-IF.
```

RecordA

Deleting records from an ordered file



Deleting records from an ordered file



Updating records in an ordered file

RecordB
RecordH
RecordK

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

```
FILE SECTION.
    TFRec
    OFRec
    NFRec
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey = OFKey
      Update OFRec with TFRec
      MOVE OFRec+ TO NFRec
      WRITE NFRec
      READ TF
  READ OF
ELSE
      MOVE OFRec TO NFRec
      WRITE NFRec
 READ OF
END-IF.
```

Updating records in an ordered file

RecordB
RecordH
RecordK

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

```
FILE SECTION.
    RecordB
    RecordA
    RecordA
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey = OFKey
        Update OFRec with TFRec
        MOVE OFRec+ TO NFRec
        WRITE NFRec
        READ TF
  READ OF
ELSE
        MOVE OFRec TO NFRec
        WRITE NFRec
 READ OF
END-IF.
```

RecordA

Updating records in an ordered file

RecordB
RecordH
RecordK

```
FILE SECTION.
    RecordB
    RecordB
    RecordB+
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey = OFKey
      Update OFRec with TFRec
      MOVE OFRec+ TO NFRec
      WRITE NFRec
      READ TF
  READ OF
ELSE
      MOVE OFRec TO NFRec
      WRITE NFRec
 READ OF
END-IF.
```

RecordA
RecordB+

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

Updating records in an ordered file

RecordB
RecordH
RecordK

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

```
FILE SECTION.
    RecordH
    RecordG
    RecordG
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey = OFKey
      Update OFRec with TFRec
      MOVE OFRec+ TO NFRec
      WRITE NFRec
      READ TF
  READ OF
ELSE
      MOVE OFRec TO NFRec
      WRITE NFRec
 READ OF
END-IF.
```

RecordA
RecordB+
RecordG

Inserting records into an ordered file

RecordC
RecordF
RecordP

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

```
FILE SECTION.
        TFRec
        OFRec
        NFRec
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey < OFKey
      MOVE TFRec TO
  NFRec
      WRITE NFRec
      READ TF
ELSE
      MOVE OFRec TO
  NFRec
      WRITE NFRec
 READ OF
END-IF.
```

Inserting records into an ordered file

RecordC
RecordF
RecordP

```
FILE SECTION.
   RecordC
   RecordA
   RecordA
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey < OFKey
      MOVE TFRec TO
  NFRec
      WRITE NFRec
      READ TF
ELSE
      MOVE OFRec TO
  NFRec
      WRITE NFRec
      READ OF
END-IF.
```

RecordA

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

Inserting records into an ordered file

RecordC
RecordF
RecordP

```
FILE SECTION.
    RecordC
    RecordB
    RecordB
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey < OFKey
      MOVE TFRec TO
  NFRec
      WRITE NFRec
      READ TF
ELSE
      MOVE OFRec TO
  NFRec
      WRITE NFRec
      READ OF
END-IF.
```

RecordA
RecordB

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

Inserting records into an ordered file

RecordC
RecordF
RecordP

```
FILE SECTION.
    RecordC
    RecordG
    RecordC
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey < OFKey
      MOVE TFRec TO
  NFRec
      WRITE NFRec
      READ TF
ELSE
      MOVE OFRec TO
  NFRec
      WRITE NFRec
      READ OF
END-IF.
```

RecordA
RecordB
RecordC

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

Inserting records into an ordered file

RecordC
RecordF
RecordP

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

```
FILE SECTION.
    RecordF
    RecordG
    RecordF
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey < OFKey
      MOVE TFRec TO
  NFRec
      WRITE NFRec
      READ TF
ELSE
      MOVE OFRec TO
  NFRec
      WRITE NFRec
      READ OF
END-IF.
```

RecordA
RecordB
RecordC
RecordF

Inserting records into an ordered file

RecordC
RecordF
RecordP

RecordA
RecordB
RecordG
RecordH
RecordK
RecordM
RecordN

```
FILE SECTION.
    RecordP
    RecordG
    RecordG
PROCEDURE DIVISION.
OPEN INPUT TF.
OPEN INPUT OF
OPEN OUTPUT NF.
READ TF.
READ OF.
IF TFKey < OFKey
      MOVE TFRec TO
  NFRec
      WRITE NFRec
      READ TF
ELSE
      MOVE OFRec TO
  NFRec
      WRITE NFRec
      READ OF
END-IF.
```

RecordA
RecordB
RecordC
RecordF
RecordG